INTERNATIONAL MONETARY FUND

REPUBLIC OF UZBEKISTAN

STAFF REPORT FOR THE 2019 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund and the International Development Association.

Uzbekistan: Joint Bank-Fund Debt Sustainability Analysis

Risk of External Debt Distress	Low
Overall Risk of Debt Distress	Low
Granularity in the Risk Rating	Not applicable
Application of Judgment	The risk ratings were not adjusted

Based on the Joint Bank-Fund Low-Income Country Debt Sustainability Analysis (LIC-DSA), Uzbekistan has a low risk of debt distress, with debt burden indicators below relevant thresholds in the baseline and all stress scenarios. Over the medium term, the public debt-to-GDP ratio is expected to increase moderately, while the total external debt-to-GDP ratio is expected to decline somewhat. In addition, large foreign exchange reserve buffers mitigate potential distress concerns. The debt sustainability analysis suggests that the most significant risks could result from worse-than-expected external flows (mostly lower remittances) and significantly lower exports. The government should carefully manage external borrowing to maintain Uzbekistan's strong external position.[1]

[1] Due to statistical revisions and progress of structural reforms, comparisons with previous DSAs may not be informative. In particular, significant historical revisions of national accounts and balance of payments statistics hinder comparability. Moreover, given significant progress on transition reforms, staff has adjusted some of the macroeconomic projections to fit earlier experiences of transition economies.

BACKGROUND

1. Stocks of total external and public and publicly-guaranteed (PPG) debt remain low and are expected to rise only moderately over the medium term.[2] At the end of 2018, public and publicly-guaranteed external debt amounted to 20½ percent of GDP, while private external debt stood at 14 percent of GDP (text table).[3] The relatively low levels of external debt reflect a history of targeting external and fiscal surpluses under Uzbekistan's previous state-led growth model, a policy that also aimed at building up large international reserve buffers.

Uzbekistan: External Public and Private Debt, 2018

	Millions of U.S. dollars	Percent of GDP	Percent of external debt
Total external debt	17,078	34.5	100.0
Public and publicly-guaranteed debt	10,017	20.5	58.7
Government debt	7,288	14.9	42.7
Guaranteed debt	2,729	5.6	16.0
Private debt	7,061	14.0	41.3

Sources: Uzbekistan authorities; IMF staff estimates.

Uzbekistan: Public and Publicly-Guaranteed Debt, 2018

	Millions of U.S. dollars	Percent of GDP	Percent of PPG debt
Total public and publicly-guaranteed debt	10,089	20.6	100.0
Public and publicly guaranteed debt	10,017	20.5	99.3
Domestic debt	72	0.1	0.7

Sources: Uzbekistan authorities; IMF staff estimates.

2. Almost all of Uzbekistan's public and publicly-guaranteed debt is denominated in foreign currency, with only a small share in domestic currency. At end-2018, about 55 percent of PPG debt was owed to multilateral creditors, 35 percent to bilateral creditors, four percent to commercial creditors, and less than one percent represented domestic treasury bills and bonds. After paying off all domestic currency debt in 2011, the government began again issuing treasury bills and bonds in 2018 to set benchmarks for the domestic financial sector. Moreover, in early-2019, the government for the first time issued Eurobonds for one billion U.S. dollars, which raised public external debt of the government by about 1¾ percent of GDP.

[2] PPG debt consists of debt of the general government and debt of state-owned enterprises (SOEs) guaranteed by the government. External debt of SOEs that are not guaranteed by the government are included in private external debt.

[3] The levels of both debt indicators doubled approximately following FX liberalization in September 2017, when the official FX rate converged to the much more depreciated parallel market FX rate.

3. Public debt coverage is broad. Public debt covers debt of the general government (central government, local government, the pension fund, and other extrabudgetary funds) and SOE debt guaranteed by the government (see text table). However, data limitations undermine the completeness and comprehensiveness of PPG external debt data. In addition, the amount of non-guaranteed SOE debt is currently not known. Efforts to improve debt coverage are under way. (A joint IMF/World Bank mission in April 2019 worked to improve the quality of debt data and develop a medium-term debt management strategy). Most of reported private external debt reflects joint ventures between Uzbek SOEs and other firms, mostly in the energy sector. The contingency stress tests are based on standard parameters (see text table), implying a 2 percent of GDP shock to SOE debt and a 5 percent of GDP shock in case of financial market default. Currently, there are no significant Public Private Partnership (PPP) projects in Uzbekistan.

Uzbekistan: Public Sector Coverage in DSA

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	X
3	Other elements in the general government	X
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	X
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

	The country's coverage of public debt	The general government, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0	
4	PPP	35 percent of PPP stock	0.0	No significant PPPs currently
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0	
	Total (2+3+4+5) (in percent of GDP)		7.0	

1/ The default shock of 2 percent of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.).

4. Uzbekistan's debt-carrying capacity is assessed as strong. The new debt sustainability framework for LICs uses a composite indicator (CI) to capture factors affecting a country's debt-carrying capacity. In particular, the CI uses a weighted average of the World Bank's Country Policy and Institutional Assessment (CPIA) score for Uzbekistan and the country's real GDP growth, remittances, foreign exchange reserves, and global growth. The calculation of the CI is based on 10-year averages of the variables, using 5 years of historical data and 5 years of projections. Uzbekistan's present CI score is calculated to be 3.06, which is just above the 3.05 lower bound for strong debt carrying capacity. The strong reading for the indicator largely reflects Uzbekistan's high international reserves (text table).

Uzbekistan: Composite Indicator of Debt-Carrying Capacity

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Strong	Strong	Strong	Medium
	3.06	3.06	3.43

Note: Until release of the April 2019 WEO vintage, the classification and corresponding score of the two previous vintages were based solely on the CPIA.

Source: IMF staff estimates.

Uzbekistan: Calculation of Composite Indicator of Debt-Carrying Capacity

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components (percent)
CPIA	0.39	3.50	1.35	44.0
Real growth rate (in percent)	2.72	6.45	0.18	5.7
Import coverage of reserves (in percent)	4.05	57.96	2.35	76.6
Import coverage of reserves^2 (in percent)	-3.99	33.60	-1.34	-43.7
Remittances (in percent)	2.02	2.41	0.05	1.6
World economic growth (in percent)	13.52	3.58	0.48	15.8
CI Score			3.06	100.0
CI rating			Strong	

Applicable Thresholds for Debt-Carrying Capacity

	In percent		
	Weak	Medium	Strong
External debt burden threshold			
PV of debt-to-exports	140	180	240
PV of debt-to-GDP	30	40	55
Debt service-to-exports	10	15	21
Debt service-to-revenue	14	18	23
Total public debt benchmark			
Public debt-to-GDP	35	55	70

MACROECONOMIC ASSUMPTIONS

5. The macroeconomic assumptions in this DSA are broadly unchanged compared with last year's DSA, except for the external current account outlook (text table).

- **Real growth:** Projected real GDP growth rates have been marginally upgraded for 2019-20, with growth driven by domestic demand. As reforms progress, consumption is forecast to expand, and investment is expected to remain at robust levels. Medium-term growth is projected at 6 percent, as in the previous DSA. Inflation is projected to be high in the near term as price liberalization as well as relative price and wage adjustments continue, with energy prices in particular converging to cost-recovery levels. Over the medium term, inflation gradually decline as price liberalization is completed.

- **Fiscal:** The overall fiscal deficit, which combines the consolidated fiscal deficit and the balance of the government's policy-based lending operations, is projected to stabilize at 1¾ percent of GDP over the medium term (versus 1½ percent of GDP in the 2018 DSA).

- **External:** The current account deficit is significantly larger than previously projected, reflecting a host of factors, including opening and modernization of the economy, as reflected in the surge of capital goods imports in 2018. Staff also revised the external projections to fit better the experiences of earlier transition economies. These revisions imply a higher current account deficit and a faster real appreciation than in the previous DSA. At the same time, FDI has been revised up assuming reforms continue and stimulate FDI inflows. International reserves are projected to remain relatively stable in the medium term as government borrowing and FDI cover most of the current account deficit during the transition.

Uzbekistan: Comparison of Selected Macroeconomic Indicators, 2017-21

	2017	2018	2019	2020	2021
Real GDP growth (percent)					
Current DSA	4.5	5.1	5.5	6.0	6.0
Previous DSA	5.3	5.0	5.0	5.5	6.0
Overall fiscal balance (percent of GDP) 1/					
Current DSA	-1.9	-2.1	-1.6	-1.8	-1.8
Previous DSA	-2.6	-1.0	-1.1	-1.1	-1.4
Current account (percent of GDP) 1/					
Current DSA 2/	1.2	-6.9	-6.5	-5.6	-4.7
Previous DSA	3.0	0.3	-0.8	-2.1	-2.1

Sources: Country authorities; World Bank; and IMF staff estimates.

1/ Current and previous data were adjusted for recent revisions, which increased nominal GDP by about 25 percent.

2/ Balance of payments (BOP) statistics revision in 2018 implied a current account deficit ½ percentage points higher due to the adoption of a new methodology.

REALISM TOOLS

6. The projections seem to be realistic (Figure 4). Fiscal and real sector projections are consistent according to the realism tests. The change in the primary balance over the next 3 years is close to the median of the cross-country distribution (zero). Nevertheless, for 2019, a small fiscal tightening (0.2 percent of GDP) is projected with limited impact on real GDP growth. However, as past investments and reforms spur GDP growth, the negative growth impact of the fiscal tightening will be more than offset. At the same time, the contribution of government capital expenditure on growth is projected to decline as the reforms reduce the footprint of the government in the economy.

EXTERNAL AND PUBLIC DSA

Total Public and Publicly-Guaranteed (PPG) Debt

7. PPG debt is projected to rise modestly over the next 20 years (Table 1-2). Under the baseline, the government's primary deficit will stabilize at 1½ percent of GDP (Table 2). Disbursements of new debt are assumed to average around 5 percent of GDP per year, as the government uses multilateral and bilateral official borrowing for financing investment and to support its reform plans:

- Historically, about half of PPG external borrowing came from multilateral creditors and one third from official bilateral creditors. These debts typically have maturities on the order of 20 years and implicit interest rates of about 2 percent. However, the Eurobonds issued in 2019 had maturities of five and ten years and interest rates of 4¾ and 5⅜ percent, respectively.
- For SOEs, about three-quarters of guaranteed debts reflected official bilateral creditors, with a small portion from commercial creditors. Official borrowing by SOEs has been on terms similar to that of the government. Commercial borrowing has an average maturity of about 5 years with implicit interest rates of about 2½ percent.

8. The projections assume borrowing maturities and interest rates will remain comparable to their historical values. Under these assumptions, the PPG debt-to-GDP ratio is projected to gradually rise from 20½ percent of GDP in 2018 to 26 percent of GDP in 2039.

9. Debt burden and service ratios show minimal increases under the baseline scenario (Table 4). The solvency indicator, the present value (PV) of public debt-to-GDP, increases marginally from 16 percent in 2018 to about 20 percent in 2029, which is below the 70 percent benchmark. Solvency and liquidity indicators normalized by revenue show a deterioration in 2019 due to the tax reform—which reduced the tax burden—but afterwards the indicators increase only marginally. The PV of the public debt-to-revenue ratio ticks up from 73 to 77 percent between 2019 and 2029, while the public debt service-to-revenue ratio remains below 10 percent.

10. Domestic debt remains limited. The government aims to keep issuing moderate levels of domestic debt in domestic currency to support financial sector development. Given high inflation, the cost of local debt can be high, while the government has sizable liquid assets amounting about 30 percent of GDP. As a consequence, domestic debt issuance is expected to remain negligible.

Total External Debt

11. Under the baseline scenario, total external debt declines from 34 percent of GDP in 2018 to 31 percent of GDP in 2024 and remains around that value until 2039 (Tables 1-2 and Figure 1). While public external debt increases over the next 20 years, private external debt is projected to decline. An important driver of the external debt decline is the projection of persistent real exchange appreciation, which is consistent with the experiences of earlier transition economies (reflected in the debt dynamics residuals reported in Tables 1-2).

12. Under the baseline, all PPG external debt indicators remain below their indicative thresholds (Table 3). As regards solvency indicators, the PV of PPG external debt-to-GDP rises from 18 percent in 2019 to 20 percent in 2039 and is below its indicative threshold of 55 percent throughout the period. The PV of PPG external debt-to-exports ratio would rise from 69 percent in 2019 to about 97 percent in 2039, less than half the indicative threshold of 240 percent. As regards liquidity indicators, the PPG debt service-to-exports and to-revenue ratios stay below 10 percent. Both indicators remain below their thresholds, which are respectively 21 and 23 percent.

13. The ratio of private external debt to GDP will decline as joint ventures reach the repayment stage. In many cases these are debts related to joint ventures in which SOEs are participants. The creditors of these debts are largely foreign commercial banks and corporations. Private external debt is projected to decline sharply in 2019 and 2020, as some enterprises pay off outstanding debts as investment projects are completed and they reach the debt repayment stage. Thereafter, private external debt is projected to decline gradually and stabilize at around 6 percent of GDP.

14. While total external debt is expected to remain stable going forward, the underlying drivers will change (Figure 3). In the past, the non-interest current account was in surplus and the exchange rate was the main factor driving the external debt ratio. In particular, the 2017 depreciation almost doubled the external debt-to-GDP ratio. Looking forward, productivity growth is expected to contribute to appreciation of the real exchange rate, as was observed in other earlier transition economies. Thus, moderate overall fiscal deficits (of about 2 percent of GDP) and significant current account deficits (on the order of 4-6 percent of GDP) are expected to drive debt dynamics.

Stress Testing and Risks

15. The DSA shows that debt ratios are robust to a range of adverse shocks (Figures 1 and 2)[4]:

- A one standard deviation **shock to other external flows** —which includes remittances and FDI— would have the greatest impact on the PV of external debt-to-GDP and exports ratios (Table 3). Under this shock, the PV of debt-to-GDP ratio would rise to 29 percent in 2024 and decline to 24 percent in 2039 compared to 19 percent for both years under the baseline. The PV of debt-to-exports ratio would rise to 139 percent in 2024 and decrease to 116 percent in 2029 compared to about 92 percent under the baseline. Nonetheless, all ratios would remain below the benchmark thresholds.
- A one standard deviation **shock to exports** would have a considerable impact, similar to other external flows. As in a shock to other flows, all the stock and flow ratios would remain below standard thresholds.
- A combination shock (of one-half standard deviation in GDP growth, fiscal balance, exports, financing flows, and depreciation) would have the next highest impact. But again, stock and flow ratios would remain below standard thresholds.

16. Overall, Uzbekistan's risk of external debt distress remains low. The stock of external debt is projected to decrease to about 31 percent of GDP by 2029, while the stock of overall public debt including guaranteed debt is projected to reach about 25 percent of GDP (Tables 1-2). All debt stocks and debt service ratios are projected to remain well below the relevant indicative thresholds (Figures 1-2, Tables 2-4). The DSA outlook benefits from robust growth and the continued relatively low cost of financing from concessional borrowing, which underscores the importance of policies that safeguard sustainable catch-up growth and external stability. Large fiscal buffers (about 30 percent of GDP) and sizable international reserves (about 13 months of imports) are important risk-mitigating factors.

17. Market-financing risk is low (Figure 5). Low gross financing needs and the reduced sovereign spreads support the low risk of potential liquidity needs. All debt stocks and debt service ratios are below the relevant thresholds, signaling some margin to manage debt. In addition, large liquid fiscal buffers are available to cope with temporary adverse shocks.

CONCLUSION

18. Based on the debt sustainability analysis, Uzbekistan's risk of public debt and external debt distress is low. All solvency and liquidity indicators are projected to remain below their respective thresholds under both the baseline and stress scenarios.

[4] Results of stress test of public debt and PPG external debt are similar so the results for PPG external debt are only discussed.

19. Debt sustainability ratios could worsen if external borrowing is significantly higher than projected. This analysis assumes the increase in external borrowing is modest, i.e. after an initial boost as reforms get underway, external PPG borrowing remains around 4-5 percent of GDP. Additional external borrowing could result in higher growth, exports, and revenues, but could impose an additional burden if not used wisely.

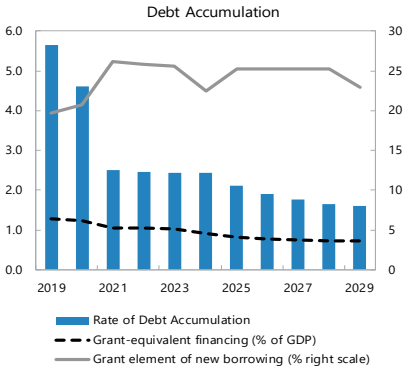
20. The authorities agree with the staff's views. They concur that the risk of debt distress is low, given the significant buffers and low debt-to-GDP ratio. The authorities share the view that investment needs must be addressed in a context of sound macroeconomic framework, including a sound fiscal policy. The authorities are also committed to strengthening debt management capacity to further minimize the risk of debt distress.

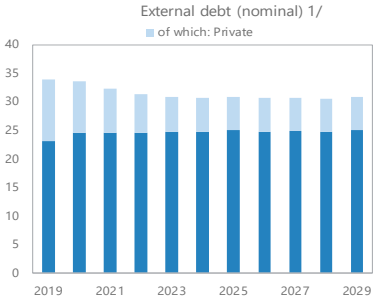
Table 1. Uzbekistan: External Debt Sustainability Framework, Baseline Scenario, 2018-39

(In percent, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**34.5**	**34.0**	**33.5**	**32.2**	**31.4**	**30.9**	**30.6**	**30.9**	**32.0**	**18.6**	**31.5**
of which: public and publicly guaranteed (PPG)	**20.5**	**23.1**	**24.6**	**24.6**	**24.6**	**24.7**	**24.8**	**25.1**	**26.1**	**9.8**	**24.6**
Change in external debt	0.4	-0.5	-0.5	-1.3	-0.8	-0.5	-0.3	0.3	0.1		
Identified net debt-creating flows	**11.0**	**3.2**	**2.0**	**1.0**	**0.5**	**0.2**	**0.1**	**0.0**	**0.0**	**-3.0**	**0.7**
Non-interest current account deficit	**6.0**	**5.5**	**4.7**	**3.9**	**3.6**	**3.6**	**3.4**	**3.4**	**3.4**	**-1.8**	**3.8**
Deficit in balance of goods and services	18.4	16.4	14.5	12.9	11.9	11.3	10.7	10.7	10.7	**3.9**	**11.9**
Exports	28.0	25.8	23.6	22.0	21.4	21.2	21.1	21.1	21.1		
Imports	46.4	42.2	38.1	34.9	33.3	32.5	31.7	31.7	31.7		
Net current transfers (negative = inflow)	-8.3	-7.2	-6.4	-5.8	-5.3	-4.9	-4.5	-4.5	-4.5	**-2.3**	**-5.1**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	-4.1	-3.7	-3.4	-3.2	-3.0	-2.8	-2.8	-2.8	-2.8	**-3.3**	**-3.0**
Net FDI (negative = inflow)	**-1.2**	**-1.7**	**-1.8**	**-2.0**	**-2.2**	**-2.4**	**-2.4**	**-2.4**	**-2.4**	**-1.8**	**-2.3**
Endogenous debt dynamics 2/	**6.2**	**-0.6**	**-0.8**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-0.9**	**-1.0**		
Contribution from nominal interest rate	1.0	1.0	0.9	0.9	0.8	0.8	0.8	0.8	0.8		
Contribution from real GDP growth	-2.0	-1.6	-1.8	-1.8	-1.7	-1.7	-1.7	-1.7	-1.8		
Contribution from price and exchange rate changes	7.2	...	...	...	...	...	...	...	...		
Residual 3/	**-10.6**	**-3.7**	**-2.5**	**-2.2**	**-1.3**	**-0.7**	**-0.4**	**0.3**	**0.1**	**5.3**	**-1.0**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**15.6**	**17.7**	**19.2**	**19.1**	**19.2**	**19.2**	**19.3**	**19.5**	**20.5**		
PV of PPG external debt-to-exports ratio	**55.6**	**68.7**	**81.6**	**86.6**	**89.5**	**90.5**	**91.5**	**92.3**	**97.5**		
PPG debt service-to-exports ratio	**3.9**	**4.6**	**6.5**	**7.5**	**7.9**	**8.1**	**8.7**	**8.8**	**10.1**		
PPG debt service-to-revenue ratio	**3.9**	**4.7**	**6.0**	**6.4**	**6.5**	**6.6**	**7.0**	**7.1**	**8.1**		
Gross external financing need (Million of U.S. dollars)	6,002	5,424	5,381	4,965	4,887	5,010	5,385	8,379	17,449		
Key macroeconomic assumptions											
Real GDP growth (in percent)	5.1	5.5	6.0	6.0	6.0	6.0	6.0	6.0	6.0	**7.1**	**6.0**
GDP deflator in US dollar terms (change in percent)	-17.4	13.6	9.4	7.4	6.1	6.2	5.7	1.5	1.0	**-2.3**	**5.8**
Effective interest rate (percent) 4/	2.7	3.3	3.1	3.0	2.9	2.8	2.8	2.7	2.6	**2.4**	**2.9**
Growth of exports of G&S (US dollar terms, in percent)	13.9	10.3	6.0	6.5	9.2	11.5	11.4	7.6	7.0	**2.3**	**9.2**
Growth of imports of G&S (US dollar terms, in percent)	42.3	8.8	4.8	4.3	7.4	9.6	9.6	7.6	7.0	**8.7**	**8.2**
Grant element of new public sector borrowing (in percent)	...	19.7	20.7	26.2	25.9	25.6	22.4	22.9	24.1	**...**	**24.0**
Government revenues (excluding grants, in percent of GDP)	27.9	25.4	25.4	25.6	25.8	25.9	26.1	26.1	26.1	**28.4**	**25.9**
Aid flows (in Million of US dollars) 5/	0.0	276.0	610.0	295.0	195.0	95.0	0.0	0.0	0.0		
Grant-equivalent financing (in percent of GDP) 6/	...	1.3	1.2	1.1	1.0	1.0	0.9	0.7	0.8	**...**	**0.9**
Grant-equivalent financing (in percent of external financing) 6/	...	19.7	20.7	26.2	25.9	25.6	22.4	22.9	24.1	**...**	**24.0**
Nominal GDP (Million of US dollars)	50,485	60,490	70,156	79,887	89,851	101,155	113,366	175,660	346,685		
Nominal dollar GDP growth	-13.2	19.8	16.0	13.9	12.5	12.6	12.1	7.6	7.0	**4.8**	**12.1**
Memorandum items:											
PV of external debt 7/	29.5	28.6	28.1	26.7	26.0	25.4	25.1	25.3	26.4		
In percent of exports	105.5	110.8	119.3	121.2	121.4	119.9	119.3	120.1	125.2		
Total external debt service-to-exports ratio	25.3	19.8	20.4	19.8	18.9	18.0	17.9	18.0	19.2		
PV of PPG external debt (in Million of US dollars)	7855.6	10704.2	13488.5	15249.8	17209.2	19398.0	21868.3	34173.9	71197.2		
(PVt-PVt-1)/GDPt-1 (in percent)		5.6	4.6	2.5	2.5	2.4	2.4	1.6	1.5		
Non-interest current account deficit that stabilizes debt ratio	5.6	6.0	5.2	5.1	4.5	4.0	3.7	3.1	3.3		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Uzbekistan: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2019-29

(In percent)

Customization of Default Settings	Size	Interactions
Tailored Tests		
Combined CLs	No	
Natural Disasters	n.a.	n.a.
Commodity Prices 2/	n.a.	n.a.
Market Financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	2.2%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	4	4

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. Stress tests with one-off breaches are also presented (if any), while these one-off breaches are deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

2/ The magnitude of shocks used for the commodity price shock stress test are based on the commodity prices outlook prepared by the IMF research department.

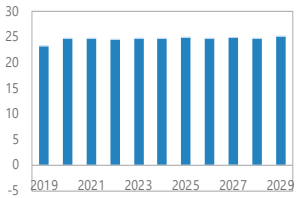
Table 2. Uzbekistan: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018-39

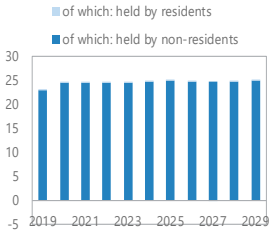
(In percent, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**20.6**	**23.2**	**24.7**	**24.7**	**24.6**	**24.7**	**24.8**	**25.2**	**27.4**	**9.9**	**24.7**
of which: external debt	**20.5**	**23.1**	**24.6**	**24.6**	**24.6**	**24.7**	**24.8**	**25.1**	**26.1**	**9.8**	**24.6**
Change in public sector debt	0.4	2.6	1.5	0.0	0.0	0.1	0.1	0.4	0.2		
Identified debt-creating flows	**-3.0**	**-1.5**	**-1.6**	**-1.1**	**-1.0**	**-0.9**	**-0.8**	**0.3**	**0.2**	**-1.5**	**-0.7**
Primary deficit	**1.6**	**1.4**	**1.5**	**1.4**	**1.5**	**1.4**	**1.4**	**1.4**	**1.4**	**-1.5**	**1.4**
Revenue and grants	27.9	25.4	25.4	25.6	25.8	25.9	26.1	26.1	26.1	**28.5**	**25.9**
of which: grants	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Primary (noninterest) expenditure	29.4	26.7	26.9	27.0	27.2	27.3	27.6	27.6	27.6	**27.0**	**27.3**
Automatic debt dynamics	**-4.5**	**-2.8**	**-3.0**	**-2.5**	**-2.4**	**-2.2**	**-2.2**	**-1.1**	**-1.2**		
Contribution from interest rate/growth differential	-1.0	-0.9	-1.2	-1.4	-1.4	-1.4	-1.4	-1.4	-1.5		
of which: contribution from average real interest rate	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0		
of which: contribution from real GDP growth	-1.0	-1.1	-1.3	-1.4	-1.4	-1.4	-1.4	-1.4	-1.5		
Contribution from real exchange rate depreciation	-3.5	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**3.4**	**2.1**	**1.3**	**0.0**	**-0.1**	**0.1**	**0.1**	**0.4**	**0.4**	**2.7**	**0.4**
Sustainability indicators	0.28	0.25	0.25	0.26	0.26	0.26	0.26	0.26	0.26		
PV of public debt-to-GDP ratio 2/	**16.2**	**18.5**	**19.7**	**19.5**	**19.4**	**19.4**	**19.5**	**20.1**	**22.4**		
PV of public debt-to-revenue and grants ratio	**58.2**	**72.7**	**77.5**	**76.2**	**75.3**	**74.9**	**74.7**	**77.0**	**85.6**		
Debt service-to-revenue and grants ratio 3/	**3.9**	**5.2**	**6.0**	**6.4**	**6.5**	**6.6**	**7.0**	**7.1**	**12.3**		
Gross financing need 4/	2.6	2.7	3.0	3.0	3.1	3.1	3.2	3.3	4.6		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	5.1	5.5	6.0	6.0	6.0	6.0	6.0	6.0	6.0	**7.1**	**6.0**
Average nominal interest rate on external debt (in percent)	2.1	2.4	2.4	2.4	2.4	2.3	2.3	2.2	2.1	**1.9**	**2.3**
Average real interest rate on domestic debt (in percent)	-20.3	77.4	-13.5	-10.5	-8.5	-7.7	-7.3	-4.2	2.3	**-12.7**	**3.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-18.0	...	...	...	...	...	...	...	...	**8.7**	**...**
Inflation rate (GDP deflator, in percent)	28.1	21.7	15.6	11.7	9.3	8.3	7.8	6.6	6.5	**16.1**	**10.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	16.7	-4.2	6.7	6.5	6.7	6.3	6.9	6.0	6.0	**8.1**	**5.4**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	1.1	-1.2	0.0	1.4	1.5	1.3	1.3	1.0	1.2	**-3.2**	**1.0**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The general government, and government-guaranteed debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

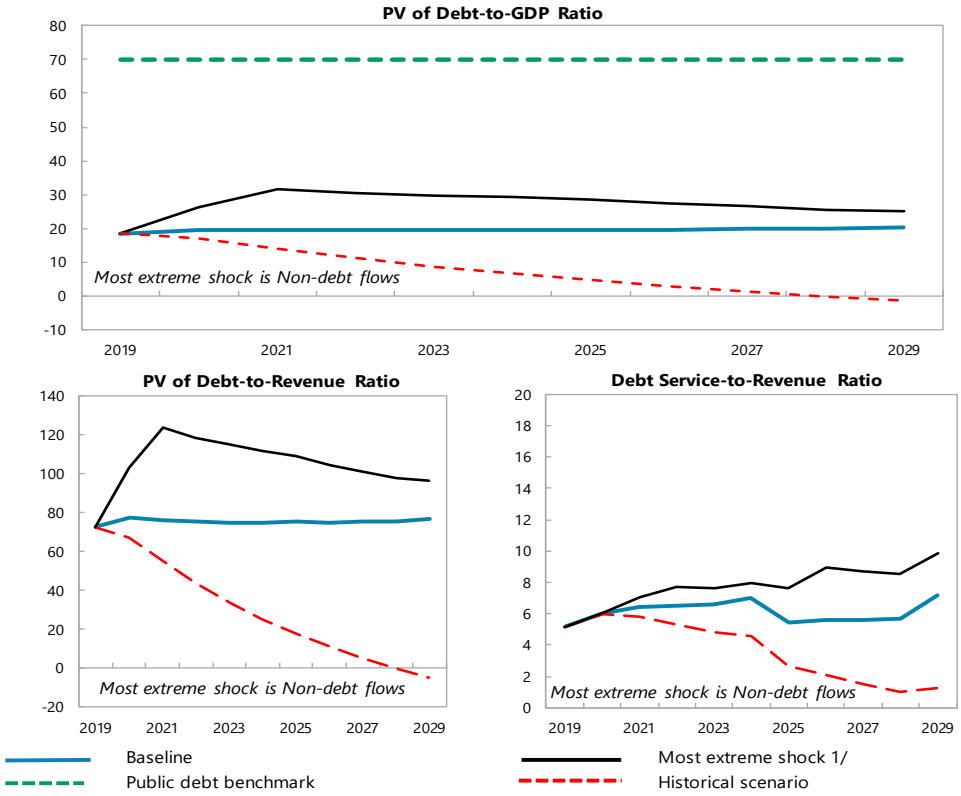
4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 2. Uzbekistan: Indicators of Public Debt, 2019-29

(In percent)



Borrowing Assumptions for Stress Tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	100%	100%
Domestic medium and long-term	0%	0%
Domestic short-term	0%	0%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.2%	2.2%
Avg. maturity (incl. grace period)	17	17
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.0%	0.0%
Avg. maturity (incl. grace period)	1	1
Avg. grace period	0	0
Domestic short-term debt		
Avg. real interest rate	-0.1%	5.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Uzbekistan: Sensitivity Analysis for Key Indicators of PPG External Debt, 2019-2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	18	19	19	19	19	19	19	19	19	19	19
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	18	16	13	11	8	6	4	1	-1	-2	-4
B. Bound Tests											
B1. Real GDP growth	18	20	20	20	20	20	20	20	20	20	20
B2. Primary balance	18	21	25	24	24	24	23	23	23	23	23
B3. Exports	18	23	28	27	27	27	26	25	25	24	24
B4. Other flows 3/	18	26	31	30	29	29	28	27	26	25	24
B5. Depreciation	18	24	21	21	21	21	22	22	22	22	23
B6. Combination of B1-B5	18	27	31	30	30	29	28	27	26	26	25
C. Tailored Tests											
C1. Combined contingent liabilities	18	25	24	24	23	23	23	23	23	23	23
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	18	21	21	21	22	22	21	21	21	21	21
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	69	82	87	89	90	92	92	92	92	92	92
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	69	68	60	50	39	28	17	7	-4	-12	-20
B. Bound Tests											
B1. Real GDP growth	69	82	87	89	90	92	92	92	92	92	92
B2. Primary balance	69	90	112	113	113	112	111	110	108	108	108
B3. Exports	69	112	174	175	174	173	169	164	160	156	154
B4. Other flows 3/	69	109	141	141	139	137	133	127	123	119	116
B5. Depreciation	69	82	75	79	80	82	83	84	85	86	87
B6. Combination of B1-B5	69	122	134	166	164	162	157	152	147	143	140
C. Tailored Tests											
C1. Combined contingent liabilities	69	105	109	111	111	110	110	109	108	107	107
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	69	82	87	91	92	93	92	91	91	91	92
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	5	7	7	8	8	9	7	7	7	7	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	5	7	8	9	9	11	6	5	3	2	4
B. Bound Tests											
B1. Real GDP growth	5	7	7	8	8	9	7	7	7	7	9
B2. Primary balance	5	7	8	9	9	10	8	9	9	9	10
B3. Exports	5	8	11	13	13	13	12	14	14	14	16
B4. Other flows 3/	5	7	8	9	9	10	9	11	11	11	12
B5. Depreciation	5	7	7	8	8	8	6	6	6	6	8
B6. Combination of B1-B5	5	7	10	11	11	12	12	13	13	13	15
C. Tailored Tests											
C1. Combined contingent liabilities	5	7	8	8	9	9	7	7	7	8	9
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	5	7	8	8	9	10	11	7	7	7	9
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	5	6	6	7	7	7	5	6	6	6	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	5	7	7	7	8	9	5	4	3	2	3
B. Bound Tests											
B1. Real GDP growth	5	6	7	7	7	7	6	6	6	6	7
B2. Primary balance	5	6	7	8	8	8	7	7	7	7	8
B3. Exports	5	6	7	8	8	8	7	8	8	8	9
B4. Other flows 3/	5	6	7	8	8	8	8	9	9	9	10
B5. Depreciation	5	7	8	8	8	8	6	6	6	6	8
B6. Combination of B1-B5	5	6	8	8	8	8	8	9	9	9	10
C. Tailored Tests											
C1. Combined contingent liabilities	5	6	7	7	7	7	6	6	6	6	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	5	6	7	7	7	8	9	6	5	5	7
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Uzbekistan: Sensitivity Analysis for Key Indicators of Public Debt, 2019-2029

(In percent)

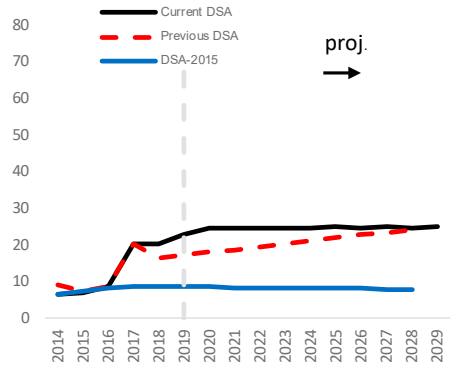
	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	18	20	20	19	19	20	20	20	20	20	20
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	18	17	14	11	9	7	5	3	1	0	-1
B. Bound Tests											
B1. Real GDP growth	18	21	22	22	23	24	25	25	26	27	28
B2. Primary balance	18	22	25	25	24	24	24	23	23	23	23
B3. Exports	18	23	27	27	26	26	26	25	24	24	24
B4. Other flows 3/	18	26	32	31	30	29	29	27	26	26	25
B5. Depreciation	18	22	20	18	16	15	14	13	12	11	10
B6. Combination of B1-B5	18	20	21	21	21	20	20	20	20	19	19
C. Tailored Tests											
C1. Combined contingent liabilities	18	25	24	24	24	23	24	23	23	23	23
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	18	20	20	20	20	20	20	20	20	20	20
Public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	73	77	76	75	75	75	76	75	76	76	77
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	73	67	55	44	34	25	18	11	5	0	-5
B. Bound Tests											
B1. Real GDP growth	73	81	84	86	89	91	94	96	100	102	106
B2. Primary balance	73	85	98	95	93	91	91	90	89	89	89
B3. Exports	73	89	107	104	101	99	98	94	93	90	90
B4. Other flows 3/	73	103	124	119	115	112	109	104	101	98	97
B5. Depreciation	73	88	78	70	64	58	54	49	45	41	38
B6. Combination of B1-B5	73	79	83	81	79	78	78	76	75	74	74
C. Tailored Tests											
C1. Combined contingent liabilities	73	99	96	93	91	90	90	89	89	88	89
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	73	78	77	76	76	76	75	75	75	75	77
Debt Service-to-Revenue Ratio											
Baseline	5	6	6	7	7	7	5	6	6	6	7
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	5	6	6	5	5	5	3	2	2	1	1
B. Bound Tests											
B1. Real GDP growth	5	6	7	7	7	8	6	6	7	7	9
B2. Primary balance	5	6	7	8	8	8	7	7	7	7	8
B3. Exports	5	6	7	7	7	8	7	8	8	8	9
B4. Other flows 3/	5	6	7	8	8	8	8	9	9	9	10
B5. Depreciation	5	7	8	8	8	8	6	6	5	5	7
B6. Combination of B1-B5	5	6	8	8	8	8	7	7	6	6	8
C. Tailored Tests											
C1. Combined contingent liabilities	5	6	7	7	7	7	6	6	6	6	7
C2. Natural disaster	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	5	6	7	7	7	8	9	6	5	5	7

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

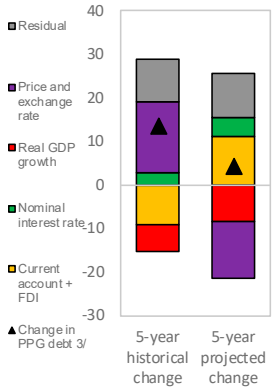
Figure 3. Uzbekistan: Drivers of Debt Dynamics – Baseline Scenario

External debt

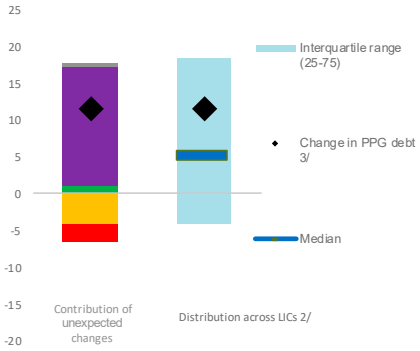
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

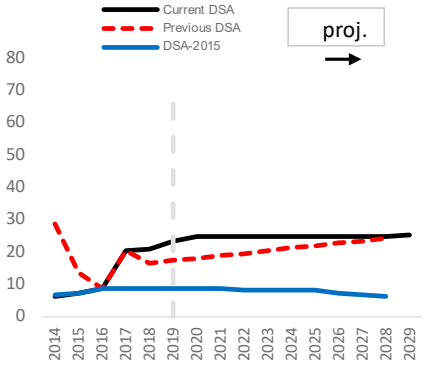


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

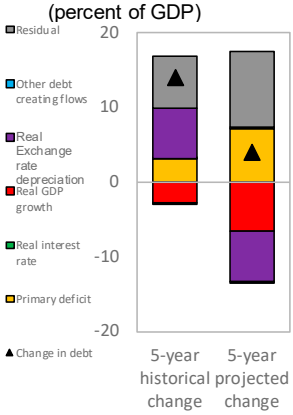


Public debt

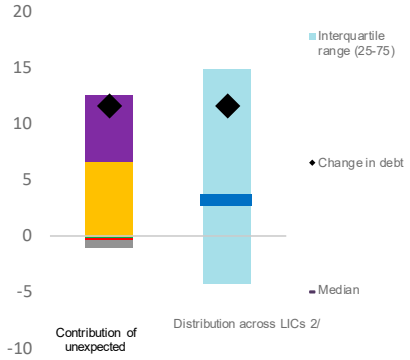
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

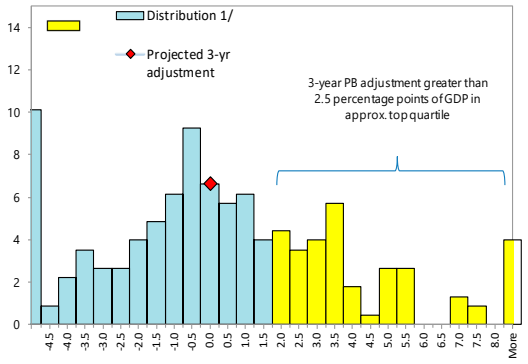
2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Source: IMF staff estimates.

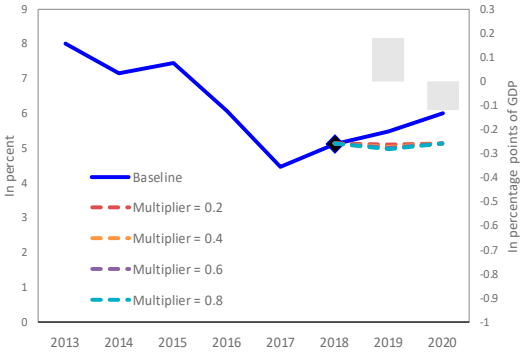
Figure 4. Uzbekistan: Realism Tools

3-Year Adjustment in Primary Balance (Percentage points of GDP)



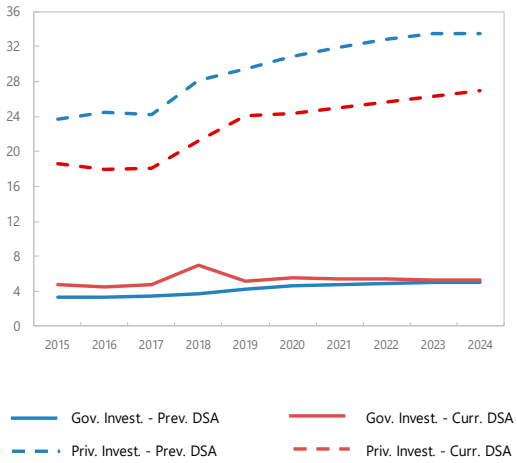
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

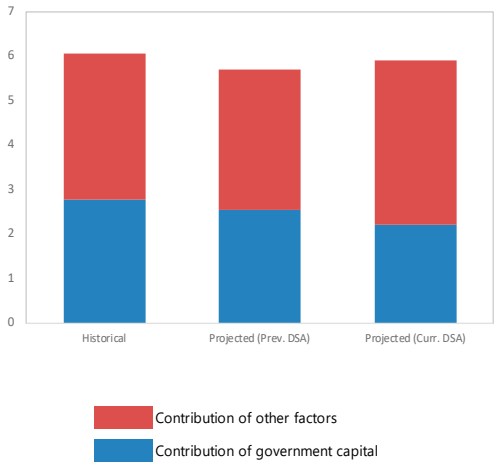


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)

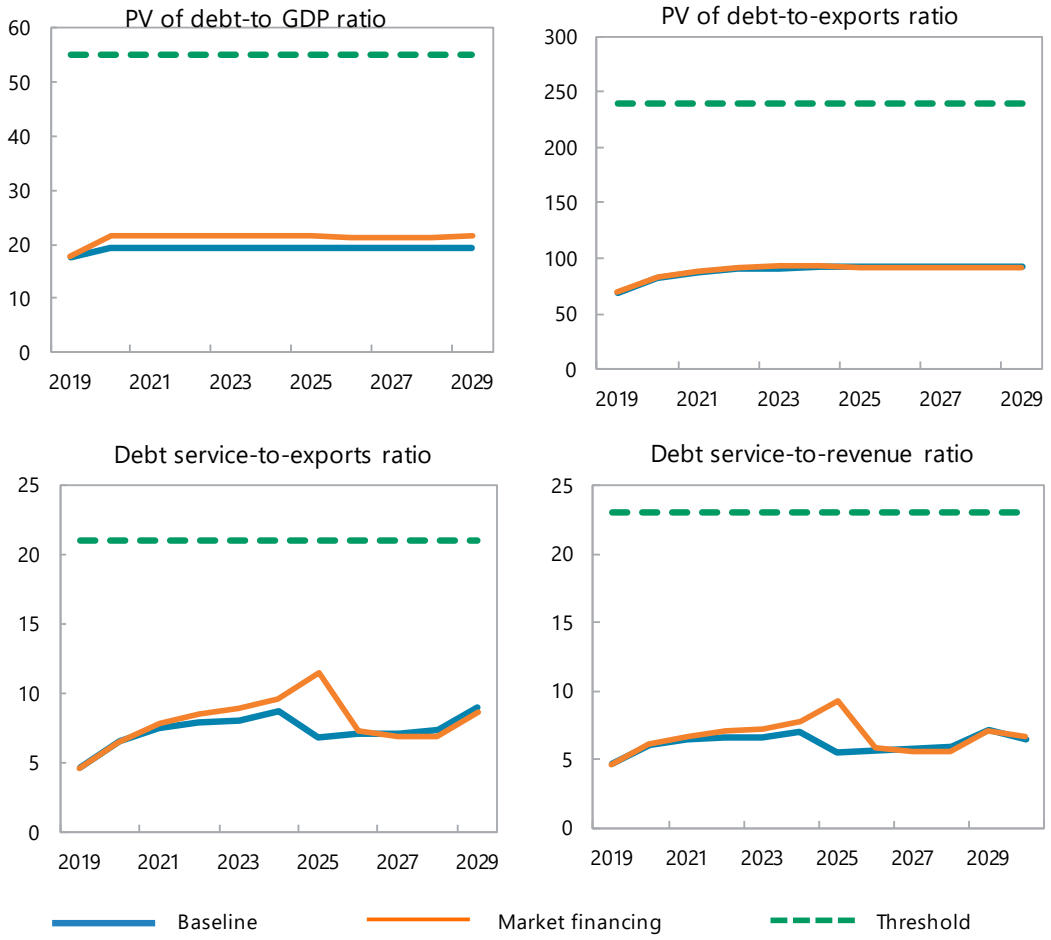


Source: IMF staff estimates.

Figure 5. Market-Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	3	250
Breach of benchmark	No	No
Potential heightened liquidity needs	Low	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ Uzbeksitan is not included in EMBI. Spread from the Uzbek 2019 international issuance.



Source: IMF staff estimates.